Problem statement

Project 73: Smart Energy Conservation Game

The problem

A household usually meets its electricity use once a month, as a bill. By then the choices behind it are hours or weeks old: how long the air conditioner ran, whether the washing was done at noon or at midnight, whether the television stayed on. Nobody gets a chance to see what a single choice cost in energy, carbon or money while there is still time to change it. It is also hard to tell which appliances matter most without measuring them.

What we set out to do

Build a small game, in Python and web technology, that gives the player a fixed daily energy budget and asks them to run a household's appliances through one simulated day while keeping consumption low. The game must show the score, the energy consumed, the CO_2 equivalent and the cost as the day unfolds. It should teach by letting the player make mistakes cheaply: go over the budget, see which appliance caused it, and try again.

Questions we want the data to answer

- Which appliances account for most of a household's daily energy, and how well do the game's power ratings match published figures?
- How much energy must a household give up to stay within a budget, and how much comfort does it lose in return?
- When energy use cannot be reduced further, how much does moving it to cheaper hours save?
- Do the answers hold if the game's own numbers are somewhat wrong?

What we deliver

A playable game with three difficulty levels and five random events; a raw and cleaned dataset (published appliance power figures and play logs); the source code at https://github.com/Pranav0774/energy-conservation-game; a final report with graphs, inferences and a sustainable-use proposal; and a short presentation.